MILITARY SPECIFICATION

TRANSPARENCY, ANIMATED, M16A1 RIFLE MAGAZINE, BOLT CARRIER AND RECEIVER GROUPS

This specification has been approved by the Naval Training Device Center, Department of the Navy. The technical requirements for procuring the animated transparency described herein shall consist of this document and the issue in effect of Specification MIL-T-82391.

1. SCOPE

1.1 This specification covers the requirements for an animated transparency of the magazine, bolt carrier and receiver groups of the 5.56-MM M16A1 rifle.

1.1.1 This item has been designated by the Naval Training Device Center as Device 3F68/1A for Navy procurement.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitations for bids or request for proposal, form a part of this specification to the extnet specified herein:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)

Military

MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-82391	Transparencies, Animated: Design and Development of, General Specification for

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

FSC 6910

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3. REQUIREMENTS

3.1 Material.- Except as specified in 3.1.1 and 3.1.2, materials used shall be in accordance with the material requirements of MIL-T-82391.

3.1.1 Metals.- All metal parts and fastenings, excluding springs, shall be of corrosion resistant steel, class 2 chromium plated metal in accordance with QQ-C-320 or Type II anodized aluminum in accordance with MIL-A-8625.

3.1.2 Wood.- Wood used shall be select grade hardwood suitable for a painted finish. The wood shall be properly seasoned, free from checks and splits and shall not be warped.

3.1.2.1 Glue.- Glue used for joining wood surfaces shall be a type not affected by moisture, lacquer, paint or sealers and shall be applied only to bare wood surfaces.

3.2 Design.- The animated transparency shall be designed to demonstrate the complete firing sequences occurring in the operation of the M16A1 rifle. The design shall be in accordance with the design requirements of MIL-T-82391 and as specified in paragraph 3.2.1 of this specification.

3.2.1 Design details.- The animated transparency shall be designed to realistically depict those components necessary to simulate the functioning of the M16A1 rifle through the eight basic steps described in the Department of the Army Field Manual (FM 23-9). The animated transparency shall be capable of demonstrating the eight basic steps in both the automatic and semiautomatic modes.

3.2.1.1 The following major components shall be depicted:

(a) A small projectable pictorial view of the entire rifle

(b) Charging handle assembly

(c) Partial barrel assembly

(d) Bolt carrier

(e) Bolt assembly

(f) Bolt cam pin

(g) Cartridge, (6 each)

(h) Magazine assembly

(i) Buffer assembly

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(j) Selector lever

(k) Automatic sear assembly

(l) Forward assist assembly

(m) Firing pin

(n) Trigger

(o) Hammer assembly

(p) Bolt catch.

3.2.1.2 The pictorial view of the rifle, as required by 3.2.1.1 (a) shall be rendered in shades of gray and that portion of the rifle demonstrated by the animated transparency shall be amber over the gray in color. The nomenclature "M16A1 RIFLE" shall be included. The view shall be provided with an opaque flap.

3.2.1.3 All operative and stationary parts of the rifle necessary to demonstrate the eight basic steps shall be depicted.

3.2.1.4 Means shall be provided for depicting the use of the forward assist assembly to chamber a round which, as a result of a malfunction, did not chamber during normal operation of the charging handle assembly.

3.2.1.5 Means shall be provided for indicating the chambered round with the bolt in a locked position. It shall not be required to demonstrate the cam action of the bolt carrier on the bolt in this animated transparency.

3.2.1.6 Means shall be provided for ejecting the round after extraction from the firing chamber. It shall not be required that ejection be performed in the same manner as in the actual weapon so long as it appears to be done by the same means.

3.2.1.7 The movement of the firing pin when struck by the hammer, shall be exaggerated.

3.2.1.8 The rounds shall be capable of being loaded in the magazine in the same manner as in the actual weapon, except that four rounds shall be considered as a fully charged magazine. The magazine shall be capable of being inserted in the magazine housing in the same manner as in the actual weapon.

3.2.1.9 Means shall be provided to prevent operation of the rifle when the selector lever is on "SAFE".

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3.2.1.10 Means shall be provided to show the action of the bolt stop held by the magazine follower after the last round is fired to hold the bolt carrier group to the rear.

3.3 Reliability.- The animated transparency shall meet the reliability requirements of MIL-T-82391.

3.4 Details of components.- Except as specified in 3.4.1 and 3.4.2, the details of components of the animated transparency shall be in accordance with the details of components requirements of MIL-T-82391.

3.4.1 Baseplate markings.- Information for orientation of the animated transparency, identification of controls or for warning notes shall be engraved with .250 inch high letters and filled with a color (black or white) contrasting with the background, by application of dry transfer lettering, by silk screening or by fluid lettering devices. Markings not engraved shall be protected by transparent sheet plastic.

3.4.2 Storage/carrying case.- The storage/carrying case for the animated transparency shall be in accordance with the storage/carrying case requirements of MIL-T-82391 except that the material for the top and bottom of the case shall be 1/4" plywood.

3.5 Construction.- Construction shall be in accordance with the construction requirements of MIL-T-82391.

3.6 Performance.- Performance shall be in accordance with the performance requirements of MIL-T-82391.

3.7 Finish and color.- Finish and color shall be in accordance with the finish and color requirements of MIL-T-82391.

3.8 Nameplates or product markings.- Nameplates or product markings shall be in accordance with the nameplates or product markings of MIL-T-82391.

3.9 Workmanship.- Workmanship shall be in accordance with the workmanship requirements of MIL-T-82391.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspections, examinations and tests.- Inspections, examinations and tests shall be in accordance with the quality assurance provisions of MIL-T-82391.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- Preservation, packag-

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ing, packing and marking shall be in accordance with the preparation for delivery requirements of MIL-T-82391.

6. NOTES

6.1 The notes of MIL-T-82391 are applicable.

CUSTODIAN:
Navy-TD

PROCURING ACTIVITY:
Navy-TD
Project No. 6910-N297

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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